[Metric]
A-A-1828B
May 13, 1996
Superseding
A-A-1828A
February 20, 1990

COMMERCIAL ITEM DESCRIPTION
LEAD, PENCILS, GRAPHITE, POLYMER

The General Services Administration has authorized the use of this commercial item description, for use by all federal agencies.

1. Scope: This Commercial Item Description covers various diameter size leads for general use on photographic, drafting film and paper with appropriate mechanical pencils.

2. Classification:

Types: I - For General Use On Photographic Film or Paper
II - For Use On Drafting Film

3. Salient Characteristics:

3.1 Type I leads shall produce uniformly dense opaque, ink-like marks when used on photographic film or paper. Marks shall be smudge resistant, reproduce clearly, and erase cleanly without ghosting when using a pencil type rubber eraser, artgun eraser, or plastic film eraser.

3.2 Type II leads shall produce dark, dense, opaque, ink-like marks when used on drafting film. Marks shall be smudge proof, reproduce clearly and erase cleanly without ghosting when using a pencil or plastic film eraser.

3.3 Leads shall produce smooth continuous markings without crumbling or chipping, and be of such strength and resilience so as to reduce breakage when used properly and shall be of the following sizes: .3 mm, .5 mm and .7 mm or as specified in the contract or order.

3.4 The dimensions shall be as follows: Length of the lead shall be 60.0 mm ± 2 mm (2.36 inches ± .08 inches) long. The diameter shall conform to Table I listed below.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to General Services Administration, Federal Supply Service, Engineering Division, 26 Federal Plaza, New York, NY 10278.

FSC 7510

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Table I

Lead Diameter (mm)	Range (mm)
0.3	0.37 to 0.39
0.5	0.55 to 0.58
0.7	0.69 to 0.73

Diameter shall be uniform throughout the entire length of the lead (see section II).

The hardness range shall be expressed as a **"grade"** as defined by ASTM C709 and be one of the following, as specified.

6H, 5H, 3 H, 2, H, F, HB, B, 2B, 3B, 4B, 5B and 6B

The hardness of 0.5 and 0.7 mm leads shall be such that no inversion shall take place as the density transmits from 6H to 4B for the leads of the same brand. With respect to the 0.3 mm leads (nominal size), no inversion shall take place as the density transmits from 6H to 2B for the leads of the same brand. The grades and lack of inversions shall be certified by the vendor to be available to the government in the form of a chart of line densities for the full range of grades, upon request.

3.5 Workmanship. Leads shall be clean and free from defects which adversely affect serviceability or performance. The container shall be in accordance with the manufacturers best commercial practice and shall have no defects that affect serviceability or appearance.

4. Regulatory Requirements: The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. Quality Assurance Requirements:

5.1 General Quality Statement. The products provided shall conform to the producer's own drawings, specifications, standards and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provision of the contract.

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5.2 Examination and Testing.

5.2.1 Suppliers are encouraged to use quality control (QC) techniques that exhibit control over their processes (e.g. (Statistical Process Control) SPC techniques) as defined in American National Standards Institute (ANSI) Z1.1, Z1.2 and Z1.3/ American Society for Quality Control (ASQC) B.1, B.2 and B.3 that systematically reduce excess variations. If used, objective evidence shall be available that demonstrates overall measurement adequacy techniques and controls. These techniques shall ensure quality levels equal to, or greater than, those cited in applicable technical document or, herein.

5.2.2 End item inspection /testing may be used by the offeror or the Government representative, as a means to determine the effectiveness of the in process quality controls. While in-process controls shall not be substituted for end item performance testing they may be used for visual and dimensional sampling.

5.2.3 For those characteristics for which there are no such controls, or for end item performance test, lot by lot sampling for inspection/test shall be required. The sample unit shall be one tube. or box of leads of either type. Sampling shall be in accordance with the American National Standards Institute (ANSI)/ American Society for Quality Control (ASQC) Z1.4, Sampling Procedures and Tables for Inspection by Attributes.

5.2.4 An inspection lot shall consist of all like items submitted for inspection at one time. The inspection level for visual, dimensional defects, and measurement testing shall be S-1 with an AQL of 4.0 percent defective. The supplier must provide objective evidence (tests and inspection records) that the presented material meets the requirements of the sampling plan as indicated above.

6. Packaging:

6.1 Preservation, packing, and marking shall be specified in the contract or order.

7. Notes:

7.2 Applicable Documents. Issues of the following documents, in effect on the date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

7.2.1 Federal Acquisition Regulation (FAR) may be obtained from the Superintendent of Documents, Government Printing Office, Washington. DC 20402.

7.2.2 Application for copies of (ANSI) Z1.1, Z1.2, Z1.3 and (ASQC) B.1, B.2 and B.3 may be obtained from the American Society for Quality Control. PO Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.

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7.2.3 Copies of appropriate ASTM's may be obtained from the American Standards for Testing Materials at 1916 Race Street, Philadelphia, PA 19103.

8.3 National Stock Number Listing (NSN). The following is a list of Nsn's assigned which correspond to this CID. The list may not be indicative of all possible nsn's associated with the CID.

7510-01-011-9456
7510-01-011-9457
7510-01-011-9458
7510-01-059-0695
7510-01-103-8266
7510-01-147-5945
7510-01-147-5947
7510-01-147-6022
7510-01-116-4059
7510-01-131-9230
7510-01-258-7517
7510-01-309-7681
7510-01-317-6421
7510-01-317-6422
7520-01-335-7935
7520-01-335-7936
7520-01-335-7937
7520-01-358-4275
7510-01-358-4276
7510-01-358-4277
7510-01-358-4278
7510-01-358-4279
7510-01-358-4280
7510-01-358-4281
7510-01-358-4282
7520-01-358-7488

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions and amendments to this Commercial Item Description until further notice.

Preparing Activity:

GSA-FSS

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